

ClimateOS

AI-Powered Climate Disaster Intelligence & Response Platform

Version: 2.0 — Hackathon MVP | **Status:** Pre-development | **Theme:** AI for Climate Change

Product Vision

Build an intelligent disaster-response layer that transforms fragmented climate data and citizen observations into verified, localized, and actionable decisions.

Core Loop: Predict → Verify → Understand → Prioritize → Act → Learn

1. Executive Summary

ClimateOS is an AI-powered disaster intelligence platform designed to help cities, authorities, responders, and citizens make faster and better decisions before, during, and after climate-related disasters.

Rather than functioning only as a prediction or alerting application, ClimateOS creates a continuous intelligence loop using climate/weather data, geographic information, historical data, citizen-generated reports, and simulated sensor inputs.

The platform converts this information into localized risk information, verified citizen incidents, dynamic risk maps, personalized emergency actions, safer evacuation routes, rescue priorities, and government response recommendations.

Key differentiator: citizens are not only recipients of alerts; they become real-time sources of ground truth.

2. Problem Statement

Existing disaster-management systems frequently address individual capabilities such as hazard prediction, emergency alerts, GIS mapping, damage assessment, evacuation planning, citizen reporting, or resource allocation. During a disaster, however, citizens and authorities still face an information gap.

Citizens need:

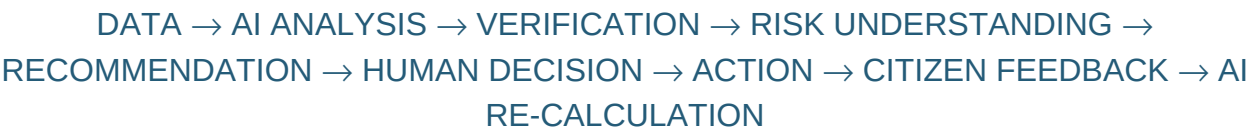
- What is happening near me?
- Am I personally at risk?
- Where should I go?
- Which route is safe?
- Which shelter is available?
- How can I request help?

Authorities need:

- Where is the disaster actually happening?
- Which reports are trustworthy?
- Where are people trapped or vulnerable?
- Which roads are blocked?
- Which areas require immediate evacuation?
- Where should limited rescue resources be deployed first?

3. Product Principle

ClimateOS does not replace emergency authorities. AI acts as an intelligence and decision-support layer, while authorized humans remain responsible for final emergency decisions.



4. Target Users

Citizens — personal risk awareness, emergency alerts, evacuation, shelter discovery, emergency reporting, and assistance requests.

Government / Disaster Authorities — city-level situational awareness, risk prediction, verification, evacuation planning, resource prioritization, and coordination.

Emergency Responders — incident priority, victim locations, route information, resource availability, and mission assignment.

5. Disaster Scope

MVP focus: Urban flooding. The architecture should later support extreme rainfall, heatwaves, cyclones, wildfires, drought, landslides, and air-quality emergencies.

For the hackathon, one disaster scenario should be demonstrated deeply rather than many scenarios superficially.

6. Core Architecture

Layer	Responsibilities
Data Sources	Weather, historical data, maps/geospatial data, citizen reports, simulated sensors
AI Fusion Core	Risk prediction, report verification, impact estimation, priority ranking, recommendations
Digital Twin	Dynamic risk map, infrastructure, hazards, incidents, routes
Government Mode	Situational awareness, recommendations, resource/rescue priorities
Citizen Mode	Personal risk, alerts, safe routes, shelters, incident reporting
Feedback Loop	Verified citizen observations update risk and response recommendations

7. Core AI Intelligence Layer

Risk Prediction Engine: calculates geographic disaster risk using weather, rainfall, historical patterns, geography, population/infrastructure information, and simulated sensors. Outputs include risk score, category, affected zone, severity, and confidence.

AI Report Verification: evaluates citizen descriptions, images, GPS, timestamp, nearby reports, weather/risk information, and available sensor information. It returns a confidence score and status such as high confidence, needs review, or low confidence.

Impact & Priority Engine: estimates urgency and ranks incidents, people, locations, and resource requirements.

Recommendation Engine: converts verified intelligence into human-reviewable actions.

8. Citizen Ground-Truth Network

Citizens can report real-world conditions that may not yet appear in official datasets.

- Flood
- Fire
- Road blockage
- Building damage
- Power failure
- Medical emergency
- People trapped
- Water shortage
- Infrastructure damage
- Other emergency

A report may contain location, photo, short description, optional voice input, incident category, and severity. AI converts it into structured disaster intelligence.

9. Dynamic Risk Map & Digital Twin

The map displays hazard zones, risk levels, citizen incidents, verified incidents, roads, blocked roads, hospitals, shelters, rescue resources, and population-risk areas.

The Digital Twin should remain lightweight for the MVP: a 2D/2.5D geospatial representation is sufficient. The objective is decision intelligence, not photorealistic 3D.

10. Personalized Risk & Action Engine

ClimateOS translates city-level risk into individual-level guidance using user location, hazard state, risk level, shelter availability, road status, and user-selected preferences.

Example: “You are inside a high-risk zone. Shelter B is 800 m away. Route C currently has the lowest estimated risk.”

11. Dynamic Safe-Route Engine

Routes are evaluated using distance, hazard severity, road blockage, flood probability, accessibility, current incidents, and emergency accessibility. The safest route is not necessarily the shortest route.

12. AI Rescue & Resource Prioritization

ClimateOS ranks incidents according to urgency and recommends where limited emergency resources should be deployed. Authorities remain responsible for accepting or rejecting recommendations.

Incident	Example condition	Priority
A	12 people, 2 injured, trapped	96
B	30 people, no injuries	72
C	Water shortage	54

13. Government Command Center

The government interface prioritizes actions rather than raw data.

- Current situation: overall city risk, active zones, critical incidents, population at risk
- AI recommendations: evacuation, rescue deployment, road closure, shelter opening, medical/relief deployment
- Prioritized incident feed
- Resource status
- Live risk map

14. Citizen Application

The citizen experience should remain simple. The home screen should surface current personal risk, active alerts, recommended action, incident reporting, and safe shelter discovery.

15. Shelter Intelligence

Shelters should include location, capacity, estimated occupancy, availability, distance, accessibility, and status. Citizens receive the nearest suitable shelter with its current status and safest route.

16. Explainable AI

Important AI outputs should show why a recommendation was made and provide a confidence score. Example factors include heavy rainfall, rising water level, historical flood patterns, low-lying terrain, and verified citizen reports.

17. Offline / Low-Connectivity Strategy

The MVP should cache last-known risk, emergency instructions, shelter information, emergency contacts, and relevant map data. The architecture should allow a future SMS/basic-communication fallback.

18. Disaster Simulation Mode

A simulation mode is a key hackathon feature. Authorities can trigger a synthetic disaster and observe risk changes, citizen reports, AI verification, route changes, and resource recommendations in real time.

19. Signature Hackathon Demo

Step	System behavior
1. Predict	Extreme rainfall increases flood risk in Zone A.
2. Risk rises	Simulated water-level input pushes Zone A into high risk.
3. Citizens report	Flooded road, trapped people, and blocked bridge reports arrive.
4. Verify	AI classifies incidents and assigns confidence.
5. Update	Digital Twin/risk map changes.
6. Recommend	Evacuate Zone A; deploy rescue team; close unsafe road; use available shelter.
7. Personalize	Citizens receive location-specific action and safe route.
8. Ground truth	A citizen reports that the recommended route is now flooded.
9. Adapt	The route is marked unsafe and the system recalculates.
10. Learn	Updated ground truth feeds the next decision cycle.

20. Functional Requirements

- FR-01 — Calculate disaster risk for geographic zones.
- FR-02 — Display risk levels on an interactive map.
- FR-03 — Allow citizens to submit location-based disaster reports.
- FR-04 — Classify reports and assign confidence scores.
- FR-05 — Rank incidents according to severity and urgency.
- FR-06 — Provide location-based personalized alerts.
- FR-07 — Recommend routes based on disaster risk.
- FR-08 — Help citizens find available shelters.
- FR-09 — Recommend emergency resource priorities.
- FR-10 — Provide a government command center.
- FR-11 — Update risk and recommendations when new information arrives.
- FR-12 — Explain major factors behind important AI outputs.
- FR-13 — Provide a disaster simulation mode.
- FR-14 — Feed verified citizen reports back into the intelligence loop.

21. Non-Functional Requirements

- **Performance:** keep the client lightweight; heavy AI processing runs server-side.
- **Scalability:** support multiple zones, citizen reports, disaster types, and authorities.
- **Reliability:** preserve critical information during degraded connectivity.
- **Security:** protect location, personal information, reports, and government data.
- **Explainability:** expose confidence and major contributing factors.
- **Accessibility:** minimize long forms and technical terminology.

22. Lightweight Technical Architecture

Component	Purpose
Citizen Web/Mobile	Personal risk, alerts, reporting, routes, shelters
Government Dashboard	Risk map, incidents, recommendations, resources
API Layer	Authentication, data exchange, orchestration
Application Services	Reports, shelters, incidents, routes, resources
AI Services	Risk, verification, prioritization, recommendations
Data Layer	Weather, geospatial, historical, reports, simulated sensors
Digital Twin	Unified spatial representation

23. Data Sources

MVP: weather/API data, historical disaster data, geographic/map data, population/infrastructure datasets, simulated sensor data, and synthetic citizen reports.

Future: IoT sensors, river gauges, government systems, satellite feeds, emergency-service systems, and traffic systems.

24. MVP Scope

Citizen: location, personal risk, alerts, incident reporting, safe route, shelter discovery.

Government: risk map, incident feed, AI verification, priority ranking, resource recommendations, disaster simulation.

AI: risk scoring, incident classification, confidence scoring, impact estimation, priority ranking, recommendation generation.

Digital Twin: lightweight 2D/2.5D city/zone representation with risk layers, infrastructure, incidents, and routes.

25. Explicitly Out of MVP Scope

- Full-scale IoT deployment
- Production government integrations
- Complex 3D city modeling
- Drone integration
- AR/VR
- Blockchain
- Large-scale social-media monitoring
- Fully autonomous emergency decisions
- Advanced satellite-image processing
- Nationwide deployment

26. Development Priorities

Priority	Build
P0 — Core Intelligence	Risk engine; citizen reporting; AI verification; dynamic risk map; incident prioritization

P1 — Action Layer	Personalized alerts; safe route; shelter intelligence; resource allocation; government recommendations
P2 — Resilience	Offline cache; SMS fallback architecture; explainable AI; simulation mode
P3 — Future	Live IoT; satellite intelligence; government integrations; multi-disaster expansion

27. Success Metrics

- **AI:** risk prediction accuracy, incident classification accuracy, verification confidence, priority-ranking consistency.
- **Response:** time from report to authority visibility; time from verified incident to recommendation; route recalculation time.
- **Citizen:** report submission time, alert comprehension, successful route/shelter recommendation.
- **System:** dashboard response time, API response time, and simultaneous simulated incidents supported.

28. Key Differentiator

ClimateOS is not simply a weather app, disaster alert application, GIS dashboard, or prediction model. It connects these capabilities into a closed, adaptive intelligence loop.

It predicts what may happen, understands what is happening, verifies what people are reporting, prioritizes what needs attention, recommends what should happen next, and updates when ground reality changes.

29. Winning Pitch

One-line pitch: ClimateOS is an AI disaster intelligence platform that turns climate data and citizens into a real-time sensing network, then converts that intelligence into safer decisions for both authorities and individuals.

Core message: The system does not just predict the disaster. It continuously learns what is happening on the ground and adapts the response.

30. Product North Star

The ultimate goal is not to generate more alerts. It is to reduce the time between:

“Something is happening” → “The right person is taking the right action.”

31. Final Product Definition

ClimateOS — AI-powered climate disaster intelligence and response platform.

Predict — identify emerging climate risks.

Sense — collect real-world information from citizens and available data.

Verify — determine the confidence of incoming information.

Understand — maintain a continuously updated Digital Twin and risk map.

Prioritize — identify the most urgent people, locations, and resources.

Act — recommend evacuation, routes, rescue, and resource deployment.

Learn — use new citizen and environmental information to continuously update the system.

Predict → Sense → Verify → Understand → Prioritize → Act → Learn